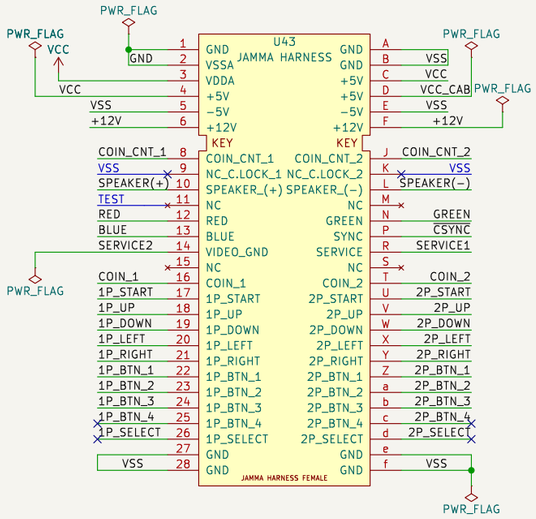
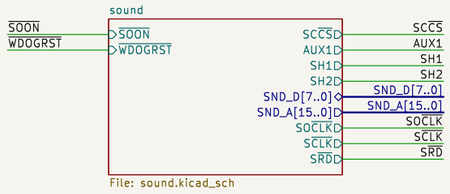
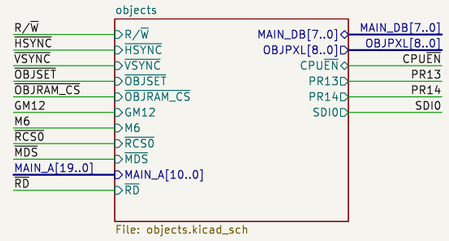
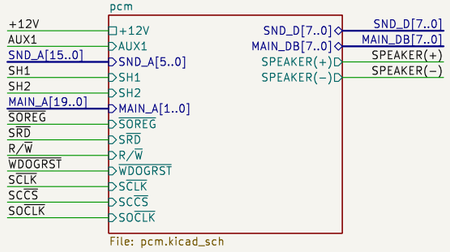
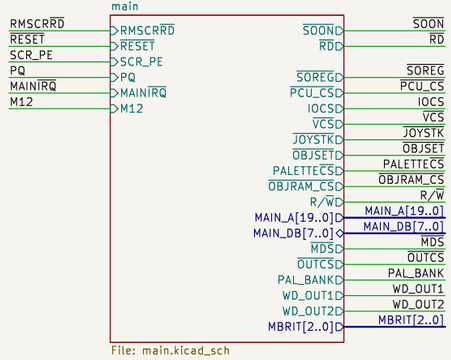
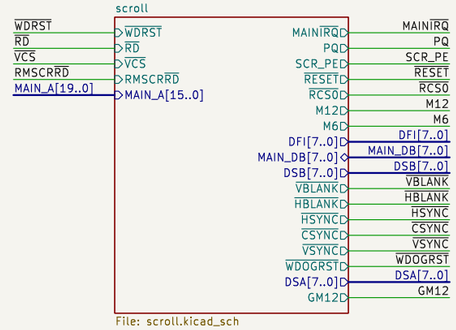


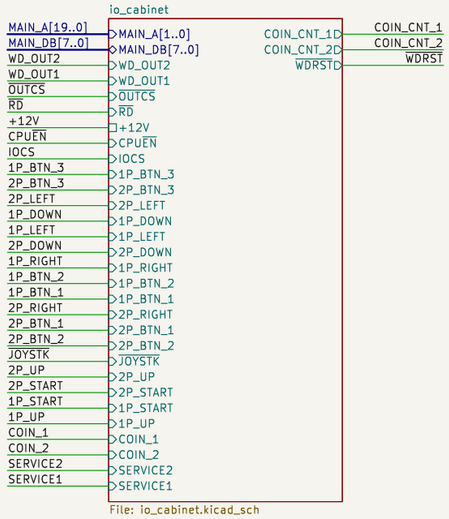



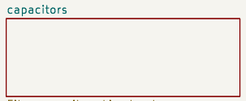



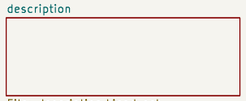



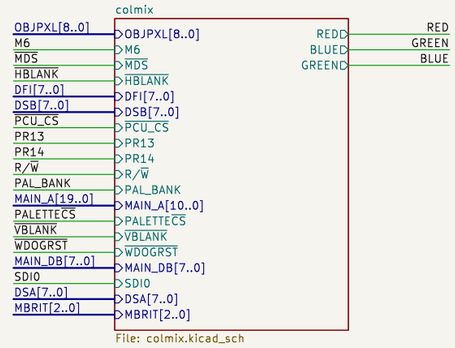




JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /
File: parodius.kicad_sch

***Title: Parodius Da!***

| | | |
|---|---|---|
| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 1/11 |

MAIN_D[7..0]

AS
SCR_PED
F19B
74LS74
PALETTECS
74LS08
F11B
DTAC
M12D
F19A
F11C
74LS08
M12
H21E
74LS04
MAINCLK
VCS
H22A
AS
VCC
4728 IAM
RA14
4.7kΩx8
Jp
JP3
HALT
PE
DTAC
FIRQ
MAINRST
MAINNMI
MAINIRQ
MPQ
MAINCLK
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D4
MAIN_D5
MAIN_D6
MAIN_D7
H12
KONAMI
053248
CPU MAIN
R/W
AS
DS
MDS
MAIN_A0
MAIN_A1
MAIN_A2
MAIN_A3
MAIN_A4
MAIN_A5
MAIN_A6
MAIN_A7
MAIN_A8
MAIN_A9
MAIN_A10
MAIN_A11
MAIN_A12
PROG_A13
PROG_A14
PROG_A15
PROG_A16
MAIN_A18
MAIN_A19
MAIN_A20
MAIN_A21
MAIN_A22
MAIN_A23
NOT CONNECTED
MAIN_A[19..0]
F5
27C010
PGM
VPP
CE
OE
GND
H5
MAIN_DB[7..0]
H7
buffers
74LS245
PROGCS2
PAGED
R/W
E16C
74LS00
A->B
CE
F10
3-StateHexBuffers
74LS367
SCR_PE
PQ
RESET
PE
MPQ
MAINRST
VSS
OEa
OEb
PWR_FLAG
PAL2
PROG_A13= PAGED ? A15 : A13
PROG_A14= PAGED ? A16 : A14
PROG_A15= PAGED ? A17 : A15
PROG_A16= PAGED ? A18 : 0
PROGCS1 = A19 & PAGED
PROGCS2 = UNPAGED & (A19 | PAGED)
PAL16L8
MAIN_A15
MAIN_A14
MAIN_A13
MAIN_A19
MAIN_A18
MAIN_A17
MAIN_A16
PAGED
MAIN_A20
PROGCS1
PROGCS2
PROG_A16
PROG_A15
PROG_A14
PROG_A13
Floating (no driver)
verified on the PCB
while running the game
Constant 1.8V observed
MBRIT[2..0]
MBRIT0
MBRIT1
MBRIT2
flip flops
74LS273
WD_OUT1
WD_OUT2
RMSCRRD
F12A
74LS32
OUTCS
R/W
PAL3
BANKR
UNPAGED
NC
PAGED
WORKCS
OBJRAM_CS
PALETTECS
BANK1
WRK
i6
i7
PAL3
o12 = i7 = 2000-2FFF when BANK1
2800-5FFF
6000-7FFF when BANKR
o13 = i6 = 3C00-3FFF
PAGED= 6000-7FFF when BANKR
8000-9FFF
PALETTE_CS= 0000-07FF when WRK
OBJRAM_CS= 2000-27FF when BANK1
WORKCS= 0000-07FF when WRK
0800-1FFF
UNPAGED= A000-FFFF (ROM PPAL)
PAL1
VCS
OBJSET
PCU_CS
MISC
IOCS
JOYSTK
PAL1
JOYSTK 3F8C-3F8F
IO_CS 3F90-3F9F
MISC 3FC0-3FCF
PCU_CS 3FB0-3FBF
OBJSET 3FA0-3FAF
VCS= AS&i7&(A8&A7&A9&A8+i6+A7)
Connected to pin 7
of alternate
YM2151 footprint.
74LS174
hex flip flop
BANKCS
74LS139
decoder
OUTCS
BANKCS
SOON
SOREG
E17
MB8464A-10W-15W
CMOS Static RAM
8K x 8 BITS
WORKCS
RD
F7B
74LS04

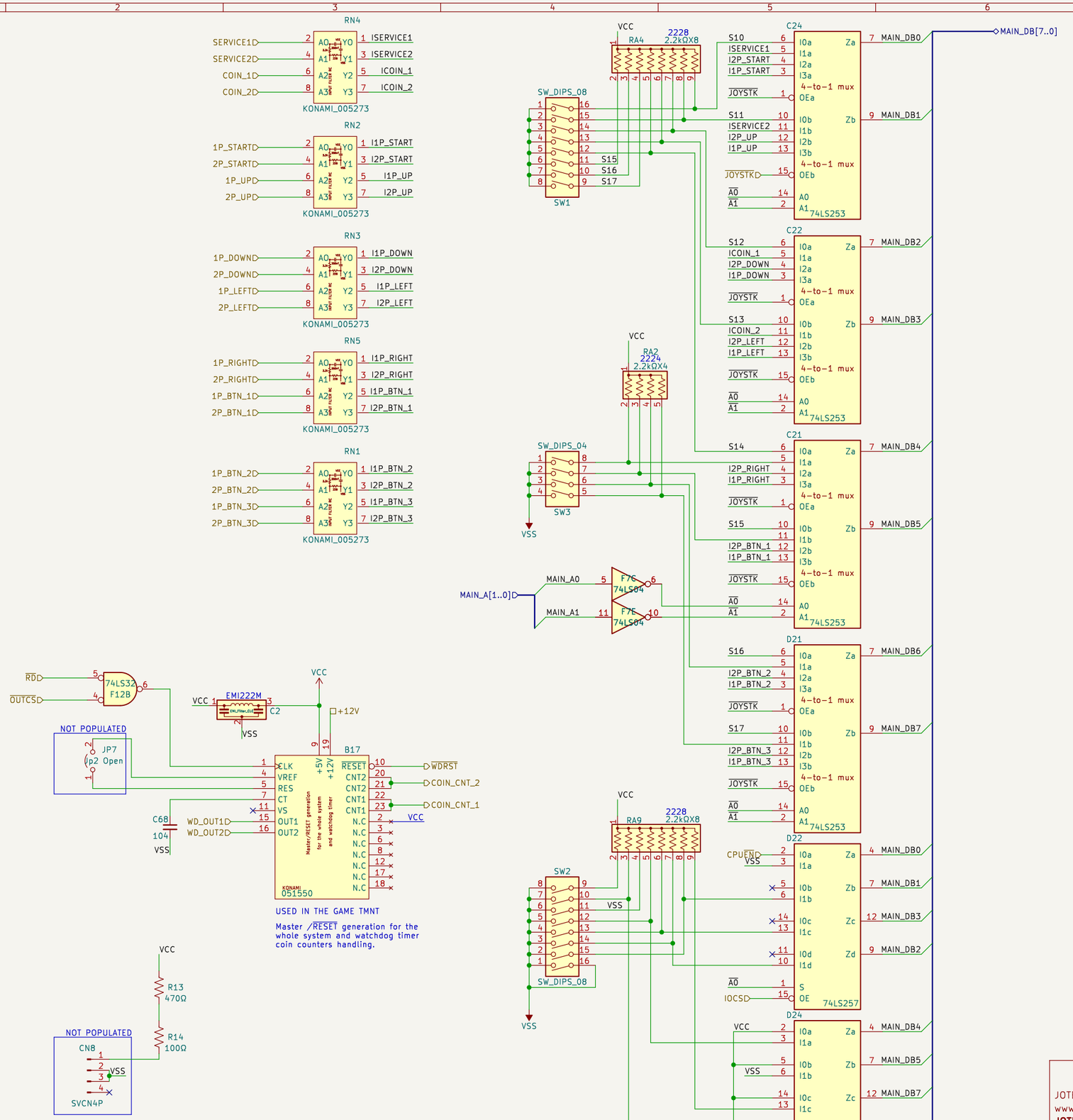
TEST HEADER
FOOTPRINT MAIN CPU
Conn_02x25_Counter_Clockwise
FP2
FP1

JOTEGO
www.patreon.com/jotego
JOTEGO

Sheet: /main/
File: main.kicad_sch

**Title: Parodius Da!**

| Size: A3 | Date: 2023-05-09 | Rev: José Tejada G. |
|---|---|---|
| KiCad E.D.A. 8.0.5 | | Id: 2/11 |




JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

***Title: Parodius Da!***

| | | |
|---|---|---|
| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 3/11 |

NOT POPULATED

RA8 4kX8 · RA13 4kX8 · RA12 4kX8 · RA10 4kX8

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7
GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15
GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23
GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMA[17..0]

CN7 — Connector_Generic_Female_First

K13 / K19 — 955D05-06-07-08 KONAMI MASK 4M - 16BIT — HN62414P — CMOS ROM

GFXROMD[31..0]

DSA[7..0] SCRA

FIX — DFI[7..0]

J15 KONAMI 051962 TILE DRAWING

MAIN_DB[7..0]

DSB[7..0] SCRB

G22 XTAL_OSC 24MHz — CP1 104 — 74LS04 H21A/H21C — H22B 74LS74 — WDRST — WDOGRST — TEST — JP5 Jp_2_Bridged

C15C 74LS07 — R18 470Ω — R20 100Ω — CSYNC

JP6 Jp_2_Bridged

SCRAMA[12..0]

H24 / J24 — MB8464A-10W-15W CMOS Static RAM 8K x 8 BITS

VD[15..0]

MAIN_A[15..0]

J20 KONAMI 052109 TILE MAP GENERATOR

TILE ATTRIBUTES · LAYER A SCROLL X · LAYER B SCROLL X · GFX ROM ADDRESS · VRAM ADDRESS · VRAM DATA · CPU DATA · GFX ROM read

MAIN_DB[7..0]

JOTEGO
www.patreon.com/jotego
JOTEGO

Sheet: /scroll/
File: scroll.kicad_sch

**Title: Parodius Da!**

Size: A3 | Date: 2023-05-09 | Rev: José Tejada G.
KiCad E.D.A. 8.0.5 | Id: 4/11

NOT POPULATED

MAIN_A[10..0]

OBJROMA[18..0]

OBJROMD[31..0]

ORAMA[10..0]

OBJROMD[7..0]

MAIN_DB[7..0]

PL[6..0]

HP[9..0]

BR[2..0]

AR[2..0]

OBJPXL[8..0]

ROM's MASK OF THE SPRITES

OBJECT ROM ADDRESS BUS

OBJECT RAM ADDRESS BUS

MAIN ADDRESS BUS

OBJECT RAM DATA BUS

GENERATOR TO DRAWING

PRIORITY CONTROL UNIT

U41
KONAMI
053245
OBJECT GENERATOR

U42
KONAMI
053244
OBJECT DRAWING

K2 / K8
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

HM6116P-3
2kB RAM

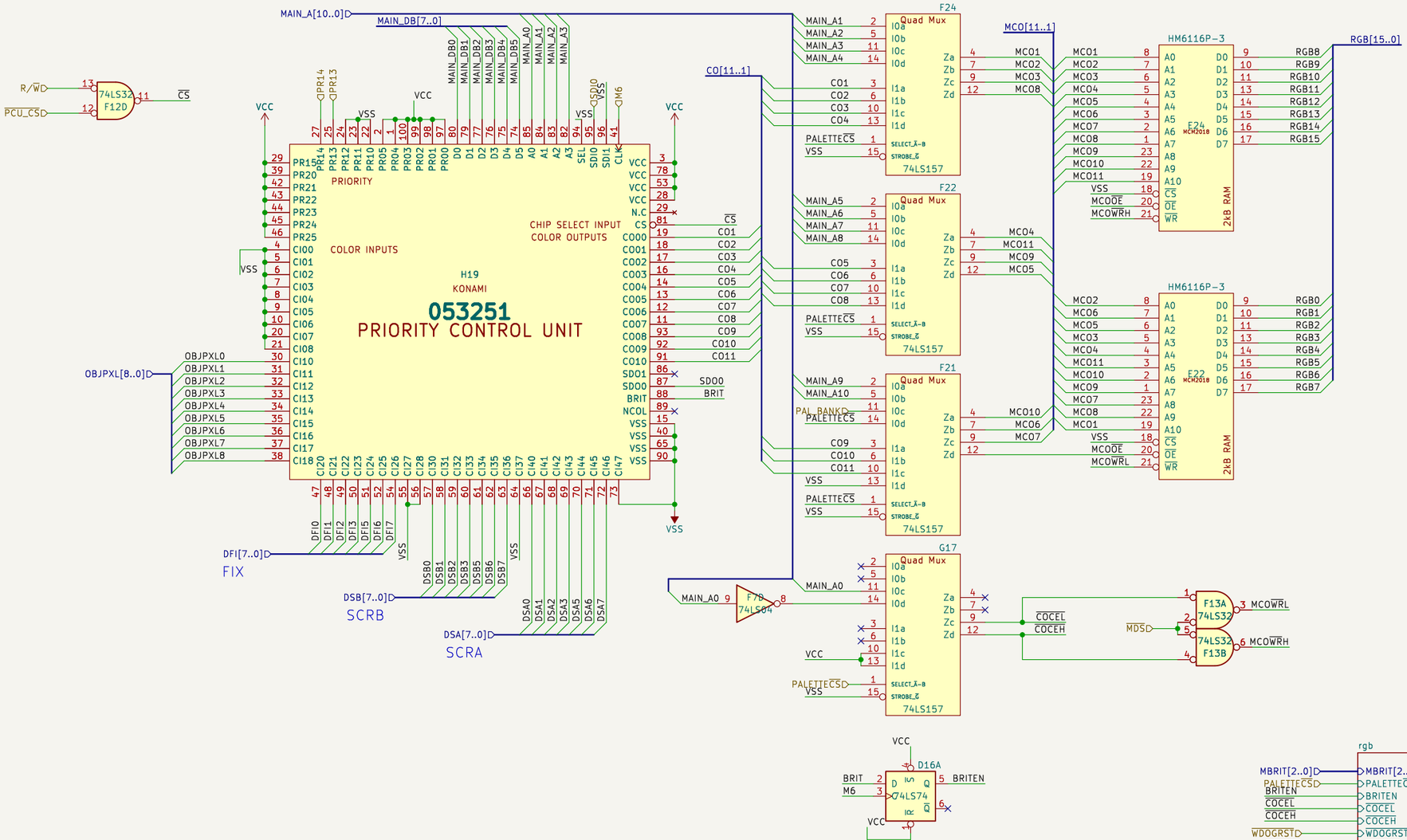
74LS245
buffers

JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /objects/
File: objects.kicad_sch

***Title: Parodius Da!***

| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada** |
|---|---|---|
| KiCad E.D.A. 8.0.5 | | Id: 5/11 |




JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /colmix/
File: colmix.kicad_sch

***Title: Parodius Da!***

| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 8.0.5 | | Id: 6/11 |

RGB[15..0]

MAIN_DB[7..0]

E21 — 74LS245 buffers (A->B)

MAIN_DB0 MAIN_DB1 MAIN_DB2 MAIN_DB3 MAIN_DB4 MAIN_DB5 MAIN_DB6 MAIN_DB7

PALETTECS

COCEL

RGB0 RGB1 RGB2 RGB3 RGB4 RGB5 RGB6 RGB7

SDOOD — C14C 74LS04

M6 VCC

D19 — 74LS273 flip flops

G3 G4 B0 B1 B2 B3 B4 SHADOW

E20 — 74LS245 buffers (A->B)

MAIN_DB0 MAIN_DB1 MAIN_DB2 MAIN_DB3 MAIN_DB4 MAIN_DB5 MAIN_DB6 MAIN_DB7

PALETTECS

COCEH

RGB8 RGB9 RGB10 RGB11 RGB12 RGB13 RGB14 RGB15

M6 VCC

D20 — 74LS273 flip flops

R0 R1 R2 R3 R4 G0 G1 G2

F20 — 74LS164 High Speed 8-Bit Serial In Parallel-Out Shift Register

VBLANK HBLANK M6 WDOGRST

AND NOT NOT

Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7

C15A 74LS07

C15B 74LS07

G4 G3 G2 G1 G0 — C17B C17D C19C C20F C19B 74LS07

B4 B3 B2 B1 B0 — C15E C15D C16D C16F C17C 74LS07

R4 R3 R2 R1 R0 — C20E C20B C20C C20A C20D 74LS07

USED IN THE GAME LETHAL ENFORCERS

S2 — KONAMI 052535 5-bit video DAC

S1 — KONAMI 052535 5-bit video DAC

S3 — KONAMI 052535 5-bit video DAC

VCC VSS

GREEN BLUE RED

SHADOW — C16B 74LS07

MBRIT2 — C17F 74LS07

MBRIT1 — C19F 74LS07

MBRIT0 — E19E 74LS07

R21 R16 R25 R15 R23 R26 R24 R22 R12 R27 R19 R17 N.M

118 Ohm

MBRIT[2..0]

MBRIT1 MBRIT0 MBRIT2

BRITEN

74LS00 E16D 74LS00 E16B E16A 74LS00

E19A 74LS07

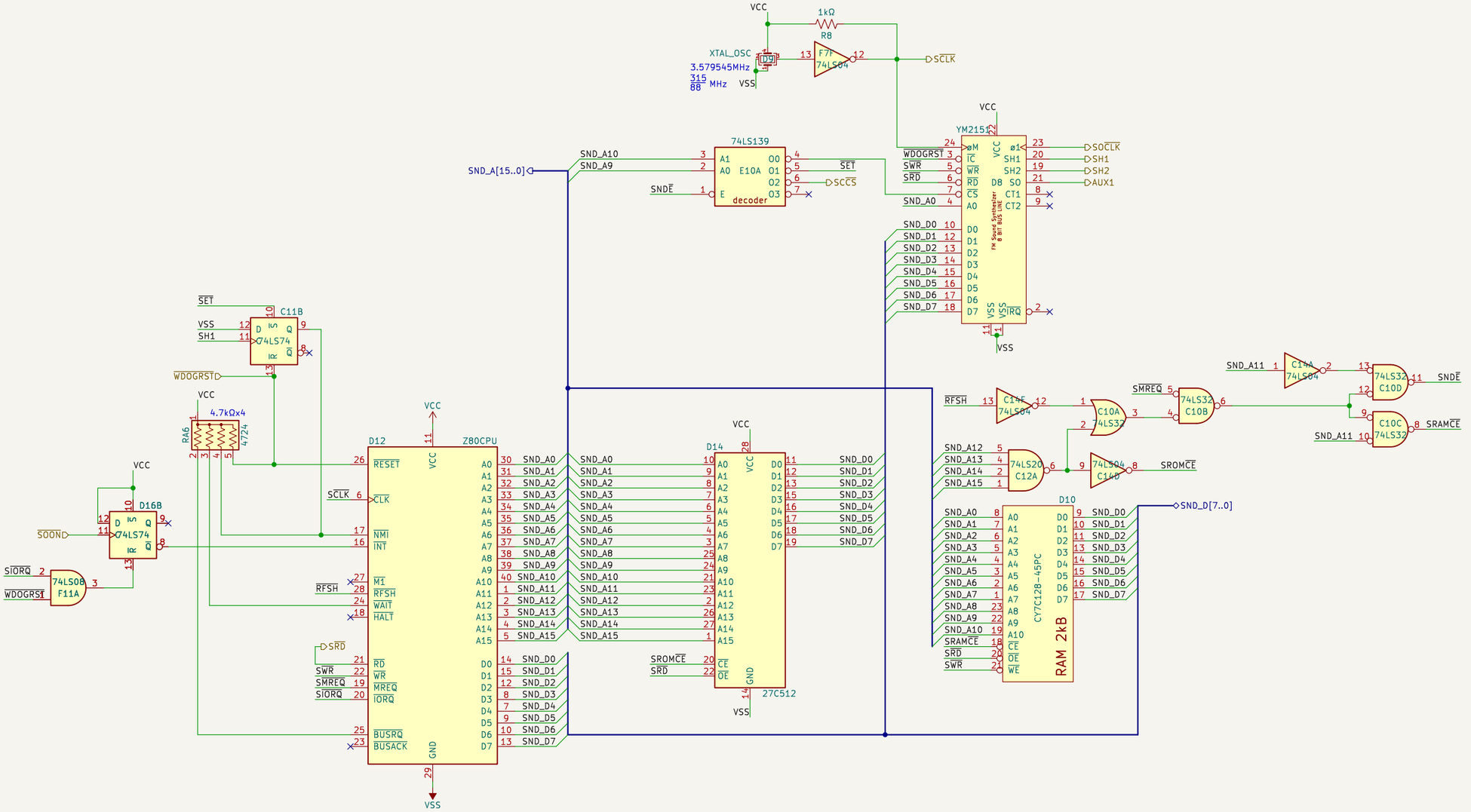
C19A C17A C19A E19F 74LS07

JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /colmix/rgb/
File: rgb.kicad_sch

***Title: Parodius Da!***

| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 8.0.5 | | Id: 7/11 |




JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /sound/
File: sound.kicad_sch

***Title: Parodius Da!***

| | | |
|---|---|---|
| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 8/11 |

NOT POPULATED

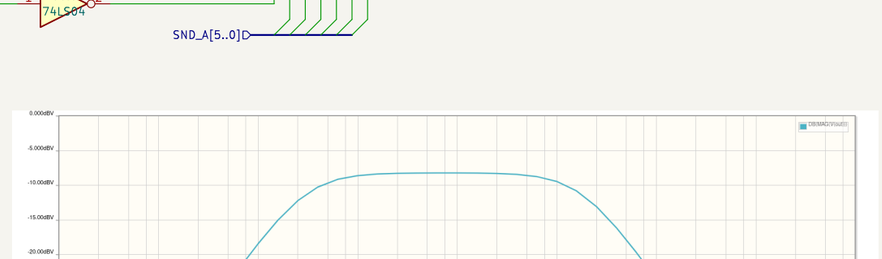
RA[19..0]

Communication between the main CPU
and the audio CPU.

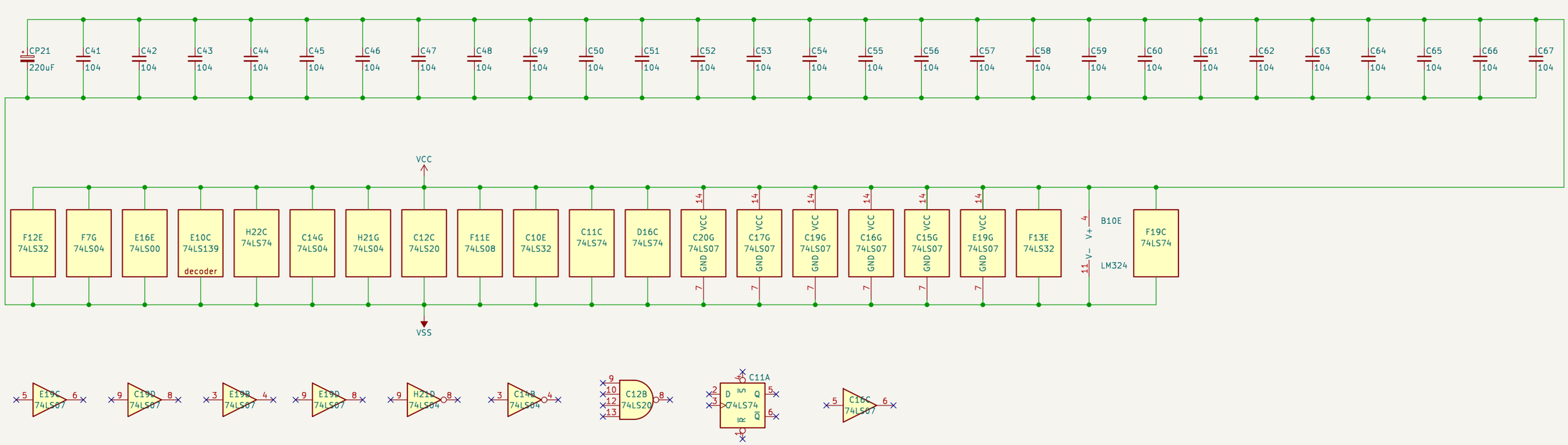
RD[7..0]

Connected to pin 21 of alternate YM2151 footprint.

JP_2_Bridged




| | | |
|---|---|---|
| JOTEGO | | |
| www.patreon.com/jotego | | |
| **JOTEGO** | | |
| Sheet: /pcm/ | | |
| File: pcm.kicad_sch | | |
| ***Title: Parodius Da!*** | | |
| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 9/11 |




JOTEGO
www.patreon.com/jotego
**JOTEGO**

Sheet: /capacitors/
File: capacitors.kicad_sch

***Title: Parodius Da!***

| | | |
|---|---|---|
| Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 10/11 |

# Parodius / Parodius DA!
# 1990

Parodius! From Myth to Laughter, released in Japan as Parodius Da!
Shinwa kara Owarai e[a] and outside Japan as Parodius
(from "Parody Gradius"), is a shoot 'em up arcade video game and the
second title in the Parodius series produced by Konami.
The European SNES version is also known as Parodius: Non-Sense Fantasy.
The gameplay is stylistically very similar to the Gradius series, but
the graphics and music are intentionally absurd.
As a result of its localised title, From Myth to Laughter is often
mistaken as the original game of the series. The lesser known original game,
Parodius: The Octopus Saves the Earth, was released for the MSX computer in
Japan.

## HARDWARE DESCRIPTION

Video Hardware : 052109 051962 053245 053244 053251
CPU : 053248
KNOWN KONAMI TMNT2 BASED GAME LIST
Detana!! Twin Bee / Bells & Whistles / Here Comes Twinbee!!
Golfing Greats
Lightning Fighters / Trigon
Parodius / Parodius DA!
Premier Soccer
Quiz Gakumon no Susume
Rollergames
Sunset Riders
Surprise Attack
Teenage Mutant Ninja Turtles : Turtles in Time

**Sequels : Gokujyou Parodius / Fantastic Journey (1994)**
**Sexy Parodius (1996)**

*Source: System 16 The Arcade Museum*
*wikipedia.org/wiki/Parodius_Da!*


JOTEGO
www.patreon.com/jotego
**JOTEGO**
Sheet: /description/
File: description.kicad_sch
***Title: Parodius Da!***
Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.**
KiCad E.D.A. 8.0.5 | Id: 11/11